Silicon (2021) 13:1871–1878
https://doi.org/10.1007/s12633-020-00570-7

ORIGINAL PAPER

Hyperbolic Two-Temperature Photo-Thermal Interaction in a Semiconductor Medium with a Cylindrical Cavity

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Received: 17 January 2020 / Accepted: 19 June 2020 / Published online: 9 July 2020
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Abstract
Photo-thermal-elastic interactions in an unbounded semiconductor media containing a cylindrical hole under a hyperbolic two-temperature are investigated using the coupled theory of thermo-elasticity and plasma waves. A new hyperbolic two-temperature model is used to study this problem. The internal surface of the cylindrical cavity is loaded by exponentially decaying pulse boundary heat flux and traction free. The Laplace transform technique are presented to obtain the exact solutions of this problem in the transformed domains by the eigenvalue approach. The inverse of Laplace transforms was numerically carried. The results show that the analytical solutions can overcome the mathematical problems to analyzes this problem. According to the numerical outcomes, the hyperbolic two-temperature thermoelastic theory offers finite velocity of the mechanical waves and thermal waves propagations.

Keywords Eigenvalues approach · Hyperbolic two-temperature · Laplace transforms · Semiconductor medium · Cylindrical cavity

PACS 52.38.Ph78.40.Fy72.20.Pa

1 Introduction

The conception of transport phenomena in solid through the expansion of spatially resolved investigation insitu that has great attentions. In this work, try to mensure the transport process established on the principle of optical beam deviations during a photo-thermal approach that can be considered as expansions of the photo-thermal deviations technique. These approaches are characterized in that it has no contact and directly produces the parameter of electronic and heat transports on the surfaces of the semiconducting or at the interface and within most of semiconducting material. Pure silicon is an actual semiconductor and is used in a wide area of the semiconductor industry, for instance, monocrystal lines Si are used to produce silicon wafer. The thermo-elasticity theory with classical two-temperatures is introduced by Gurtin and Williams [1], Gurtin and Chen [2] and Chen et al. [3] have formulated a theory of heat conduction in deformable.

bodies, which depends upon two distinct temperatures (thermodynamically temperatureT^* and conductivity temperatureϕ^*), the classic Duhem-Clausius inequality has been replaced. The first is due to the inherent mechanical process and the second is due to the thermal process between the particle and the layers of elastic materials. Youssef and El-bary [4] have presented a new hyperbolic two-temperature model in the theory of generalized thermoelasticity. It is only in the last decade was noticed, improved in several works the theories of thermo-elasticity with two temperature, and found application at most in the problem in which the discontinuity of restriction has no physical interpretation. Ezzat et al. [5] have studied the effect of dual-phase-lag heat transfer, two-temperatures and fractional time derivative model in magneto-thermo-elasticity. Ezzat and Youssef [6] have investigated the effects of ramp type heating under two-temperature model in

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three-dimensions thermo-elastic problem. Deswal and Kalkal [7] applied the state-space formulation to investigate the effect of magnetic field and initial stress with two-temperature model. Abbas et al. [8] introduced the responses of heating resources under two temperature model in a transversely isotropic thermo-elastic material without energy dissipation.

Todorovic [9, 10] have investigated the plasma, thermal and elastic wave in a semiconducting material. The changes in the propagations of plasma, thermal and elastic waves go back to the linear coupling through the thermal and the mass transports (i.e., thermal-diffusive) have included. Marin et al. [11, 12] have studied several problems for dipolar elastic materials. Song et al. [13] investigated the vibrations by the generalized thermo-elastic theory due to optically excited semiconductor and micro conductor. Othman and Marin [14] have used the Green and Naghdi theory to study the influences of heating load subjected to laser pulse on thermoelastic porous media. Yasein et al. [15] have investigated the influences of variable thermal conductivity and ramp type heating in one-dimension semiconductor medium under photo-thermal excitation. Ailawalia and Kumar [16] have studied the photothermal interaction in a semiconductor media due to ramp type heating. Alzahrani and Abbas [17] investigated the photo-thermo-elastic interactions in a semiconductor half-space without energy dissipation. Lotfy et al. [18] studied the electromagnetic and Thomson effects during the photothermal transport process of a rotator semiconducting material under hydrostatic initial stress. Lotfy et al. [19] investigated the response of electromagnetic and Thomson effects of semiconducting medium subjected to laser pulses and thermal memories during photothermal excitation. Lotfy et al. [20] have studied the effects of variable thermal conductivity of a small semiconducting cavity through under fractional-order magneto-photo-thermal model. Hobiny and Abbas [21] have studied the photothermal interactions in a two-dimensions semiconductor half-space under the GN model. Ali et al. [22] studied the reflection of waves in a rotating semiconductor nanostructure medium through torsion-free boundary condition. Jahangir et al. [23] investigated the photo-thermo-elastic interaction in a semiconductor material with two relaxation times by a focused laser beam. Many researchers [24–30] have presented several problems by using the generalized thermoelastic theories. In the domain of. Laplace, the eigenvalue method gave exact solutions without any supposed restriction on the factual physical variable.

This article aims to investigate the effect of hyperbolic two-temperature in a semiconducting media containing cylindrical cavity. The governing relations are processed by the eigenvalue approach and Laplace transform. For the considered variables, the numerical results are obtained and graphically illustrated.

2 Basic Equations

The basic equations in the context of a new hyperbolic two-temperature photo-thermal model for an isotropic, homogeneous semiconducting media in the absence of body forces heat sources and are given by [4, 31, 32]:

$$\ddot{\phi}-\ddot{T} = a\phi_{,jj}, \tag{1}$$

$$\mu u_{i,jj} + (\lambda+\mu)u_{j,ij}-\gamma_n N_{,i}-\gamma_t T_{,i} = \rho\frac{\partial^2 u_i}{\partial t^2}. \tag{2}$$

$$D_e N_{,jj}-\frac{N}{\tau}+\frac{k}{\tau}T = \frac{\partial N}{\partial t} \tag{3}$$

$$K\phi_{,jj}+\frac{E_g}{\tau}N = \rho c_e\frac{\partial T}{\partial t}+\gamma_t T_o\frac{\partial u_{j,j}}{\partial t}. \tag{4}$$

$$\sigma_{ij} = \mu\left(u_{i,j}+u_{j,i}\right)+\left(\lambda u_{k,k}-\gamma_n N-\gamma_t T\right)\delta_{ij} \tag{5}$$

where $i, j, k = 1, 2, 3$, $a > 0$ is the two-temperature parameter, u_i are the displacement components, $\phi = \phi^* - T_o$, $\varphi^*$is the variation of temperature conductivity, $N = n - n_o$, n_o is the carrier concentrations at equilibrium, $T = T^* - T_o$, $T^*$is the thermodynamic temperature. Increment, ρ is the density of material, T_o is the reference temperature,, σ_{ij} are the components of stresses, λ, μ are the Lame's constants, c_e is the specific heat at constant strain, $\gamma_n = (3\lambda+2\mu)d_n$, d_n is the electronic deformation coefficient, D_e is the carrier diffusion coefficient, τ is the lifetime of photo-generated carrier, $k = \frac{\partial n_o}{\partial T}$ is the coupling parameter of thermal activation [32], t is the time, K is the heating conductivity, $\gamma_t = (3\lambda+2\mu)\alpha_t$ and α_t is the linear thermal expansion coefficients. Let us consider a homogeneous, isotropic boundless semiconductor media containing a cylindrical cavity, whose state can be expressed in terms of the space variable r and the time t which occupying the regions $a \le r < \infty$. Due to symmetry, the component of displacement $u_r = u(r, t)$ only non-vanishing, thence the eqs. (1)–(5) are given by:

$$(\lambda+2\mu)\left(\frac{\partial^2 u}{\partial r^2}+\frac{1}{r}\frac{\partial u}{\partial r}-\frac{u}{r^2}\right)-\gamma_n\frac{\partial N}{\partial r}-\gamma_t\frac{\partial \mathrm{T}}{\partial r} = \rho\frac{\partial^2 u}{\partial t^2} \tag{6}$$

$$D_e\left(\frac{\partial^2 N}{\partial r^2}+\frac{1}{r}\frac{\partial N}{\partial r}\right) = \frac{\partial N}{\partial t}+\frac{N}{\tau}-\frac{k}{\tau}T \tag{7}$$

$$K\left(\frac{\partial^2 \phi}{\partial r^2}+\frac{1}{r}\frac{\partial \phi}{\partial r}\right) = \rho c_e\frac{\partial T}{\partial t}-\frac{E_g}{\tau}N+\gamma_t T_o\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial r}+\frac{u}{r}\right) \tag{8}$$

$$\frac{\partial^2 T}{\partial t^2} = \frac{\partial^2 \phi}{\partial t^2}-b\left(\frac{\partial^2 \phi}{\partial r^2}+\frac{1}{r}\frac{\partial \phi}{\partial r}\right) \tag{9}$$

$$\sigma_{rr} = (\lambda+2\mu)\frac{\partial u}{\partial r}+\lambda\frac{u}{r}-\gamma_n N-\gamma_t T \tag{10}$$

$$\sigma_{\theta\theta} = (\lambda+2\mu)\frac{u}{r}+\lambda\frac{\partial u}{\partial r}-\gamma_n N-\gamma_t T. \tag{11}$$

3 Application

Take into account the initial conditions and boundary conditions before solving this problem. The initial conditions are supposed to be homogeneous and are completed by the boundary on the interior cavity surface. It is without constraint subsequently the boundary conditions are written by:

$$\sigma_{rr}(a,t)=0 \tag{12}$$

the internal surface of the cavity with the heat flux [33].

$$-K\frac{\partial\phi(r,t)}{\partial r}\bigg|_{r=a}=q_o\frac{t^2e^{-\frac{t}{t_p}}}{16t_p^2} \tag{13}$$

where q_o is a constant heat flux and t_p is the pulsing heat flux characteristic time. The boundary condition of carrier density in the internal surface of the cavity can be expressed as

$$D_e\frac{\partial N(r,t)}{\partial r}\bigg|_{r=a}=s_aN(a,t) \tag{14}$$

where s_a is the velocity of recombinations on the surface.

4 Solution of the Problem

For convenience, the dimensionless physical quantities can be presented by

$$\begin{aligned}N^{\circ}&=\frac{N}{n_o},\phi^{\circ}=\frac{\phi}{T_o},T^{\circ}=\frac{T}{T_o},\left(r^{\circ},u^{\circ}\right)\\&=\eta c(r,u),\left(\sigma_{rr}^{\circ},\sigma_{\theta\theta}^{\circ}\right)=\frac{(\sigma_{rr},\sigma_{\theta\theta})}{\lambda+2\mu}\ \left(t^{\circ},\tau^{\circ},t_p^{\circ}\right)\\&=\eta c^2\left(t,\tau,t_p\right),q_o^{\circ}=\frac{q_o}{\eta cT_oK},\ \ b^{\circ}=\frac{b}{c^2}\end{aligned} \tag{15}$$

where $\eta=\frac{\rho c_e}{K}$ and $c^2=\frac{\lambda+2\mu}{\rho}$.

In these forms of the variables in eqs. (15), the basic equations are written by (for convenience, the superscript ° have been neglected)

$$\frac{\partial^2u}{\partial r^2}+\frac{1}{r}\frac{\partial u}{\partial r}-\frac{u}{r^2}-f_1\frac{\partial N}{\partial r}-f_2\frac{\partial T}{\partial r}=\frac{\partial^2u}{\partial t^2} \tag{16}$$

$$\frac{\partial^2N}{\partial r^2}+\frac{1}{r}\frac{\partial N}{\partial r}=f_3\frac{\partial N}{\partial t}+\frac{f_3}{\tau}N-\frac{f_4}{\tau}T \tag{17}$$

$$\frac{\partial^2\phi}{\partial r^2}+\frac{1}{r}\frac{\partial\phi}{\partial r}=\frac{\partial T}{\partial t}-\frac{f_5}{\tau}N+f_6\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial r}+\frac{u}{r}\right). \tag{18}$$

$$\frac{\partial^2T}{\partial t^2}=\frac{\partial^2\phi}{\partial t^2}-b\left(\frac{\partial^2\phi}{\partial r^2}+\frac{1}{r}\frac{\partial\phi}{\partial r}\right) \tag{19}$$

$$\sigma_{rr}=\frac{\partial u}{\partial r}+f_7\frac{u}{r}-f_1N-f_2T \tag{20}$$

$$\sigma_{\theta\theta}=f_7\frac{\partial u}{\partial r}+\frac{u}{r}-f_1N-f_2T \tag{21}$$

$$\begin{aligned}\sigma_{rr}(a,t)=0,&\frac{\partial N(r,t)}{\partial r}\bigg|_{r=a}=f_8N(a,t),\frac{\partial\phi(r,t)}{\partial r}\bigg|_{r=a}\\&=-q_o\frac{t^2e^{-\frac{t}{t_p}}}{16t_p^2}\end{aligned} \tag{22}$$

where $f_1=n_o\frac{\gamma_n}{\lambda+2\mu}$, $f_2=T_o\frac{\gamma_t}{\lambda+2\mu}$, $f_3=\frac{1}{\eta D_e}$, $f_4=\frac{kT_o}{n_o\eta D_e}$, $f_5=n_o\frac{E_g}{\rho c_e}T_o$, $f_6=\frac{\gamma_t}{\rho c_e}$, $f_7=\frac{\lambda}{\lambda+2\mu}$, $f_8=\frac{S_a}{\eta cD_e}$.

For $\Omega(r,t)$ function Laplace transform was written as

$$\overline{\Omega}(r,p)=L[\Omega(r,t)]=\int_0^{\infty}\Omega(r,t)e^{-pt}dt,p>0 \tag{23}$$

where p is the parameter of Laplace transforms. Hence, the governing equations can be expressed by following form

$$\frac{d^2\overline{u}}{dr^2}+\frac{1}{r}\frac{d\overline{u}}{dr}-\frac{\overline{u}}{r^2}-f_1\frac{d\overline{N}}{dr}-f_2\frac{d\overline{T}}{dr}=p^2\overline{u} \tag{24}$$

$$\frac{d^2\overline{N}}{dr^2}+\frac{1}{r}\frac{d\overline{N}}{dr}=f_3\left(p+\frac{1}{\tau}\right)\overline{N}-\frac{f_4}{\tau}\overline{T} \tag{25}$$

$$\frac{d^2\overline{\phi}}{dr^2}+\frac{1}{r}\frac{d\overline{\phi}}{dr}=p\overline{T}-\frac{f_5}{\tau}\overline{N}+pf_6\left(\frac{d\overline{u}}{dr}+\frac{\overline{u}}{r}\right) \tag{26}$$

$$\overline{T}=\overline{\phi}-\frac{b}{p^2}\left(\frac{d^2\overline{\phi}}{dr^2}+\frac{1}{r}\frac{d\overline{\phi}}{dr}\right) \tag{27}$$

$$\overline{\sigma}_{rr}=\frac{d\overline{u}}{dr}+f_7\frac{\overline{u}}{r}-f_1\overline{N}-f_2\overline{T} \tag{28}$$

$$\overline{\sigma}_{\theta\theta}=f_7\frac{d\overline{u}}{dr}+\frac{\overline{u}}{r}-f_1\overline{N}-f_2\overline{T} \tag{29}$$

$$\begin{aligned}\overline{\sigma}_{rr}(a,r)=0,\ \ &\frac{d\overline{N}(r,t)}{dr}\bigg|_{r=a}=f_8\overline{N}(a,t),\ \ \frac{d\overline{\phi}(r,t)}{dr}\bigg|_{r=a}\\&=\frac{-q_ot_p}{8\left(pt_p+1\right)^3}\end{aligned} \tag{30}$$

Differentiating eqs. (24), (25) and (26) with respect to r and using eq. (27), which can be expressed by

$$\frac{d^2\overline{u}}{dr^2}+\frac{1}{r}\frac{d\overline{u}}{dr}-\frac{\overline{u}}{r^2}=u_1\overline{u}+u_2\frac{d\overline{N}}{dr}+u_3\frac{d\overline{\phi}}{dr} \tag{31}$$

$$\begin{aligned}&\frac{d^2}{dr^2}\left(\frac{d\overline{N}}{dr}\right)+\frac{1}{r}\frac{d}{dr}\left(\frac{d\overline{N}}{dr}\right)-\frac{1}{r^2}\left(\frac{d\overline{N}}{dr}\right)\\&=n_1\overline{u}+n_2\frac{d\overline{\mathrm{N}}}{dr}+n_3\frac{d\overline{\phi}}{dr}\end{aligned} \tag{32}$$

$$\frac{d^2}{dr^2}\left(\frac{d\overline{\phi}}{dr}\right)+\frac{1}{r}\frac{d}{dr}\left(\frac{d\overline{\phi}}{dr}\right)-\frac{1}{r^2}\left(\frac{d\overline{\phi}}{dr}\right)$$

$$=t_1\overline{u}+t_2\frac{d\overline{N}}{dr}+t_3\frac{d\overline{\phi}}{dr} \tag{33}$$

where $t_1=\frac{f_6p^3}{\left(1+\frac{b}{p^2}(p+pf_6\,f_2)\right)}$ $f_2)),$ $t_2=(pf_6$ $\frac{f_1-\frac{f_5}{\tau})}{\left(1+\frac{b}{p^2}(p+pf_6\,f_2)\right)}$, $t_3=(p+pf_6\frac{f_2)}{\tau}\left(1+\frac{b}{p^2}(p+pf_6\,f_2)\right)$,

$u_1=(p^2-f_2\,t_1\frac{b}{p^2}),\ u_2=(f_1-f_2\,t_2\frac{b}{p^2}),\ u_3=f_2\left(1-\frac{b}{p^2}t_3.\right)$

$n_1=\frac{f_4}{\tau}\frac{b}{p^2}t_1,\qquad n_2=\left(f_3\left(p+\frac{1}{\tau}\right)\right.\qquad +\frac{f_4}{\tau}\frac{b}{p^2}t_2),$

$n_3=\frac{f_4}{\tau}\left(\frac{b}{p^2}t_3-1\right).$

Now to obtain the analytical solutions of the coupled eqs. (31), (32) and (33) using the eigenvalue approach proposed at references [34–37]. From eqs. (31–33), the vectors-matrices can be written in the following forms

$$DV=AV \tag{34}$$

where $=\frac{d^2}{dr^2}+\frac{1}{r}\frac{d}{dr}-\frac{1}{r^2}$, $\mathrm{V}=\left[\overline{u}\ \ \frac{d\overline{N}}{dr}\ \ \frac{d\overline{\Phi}}{dr}\right]^T$ and $A=[u_1\ \ u_2\ \ u_3\ \ n_1n_2n_3\ \ t_1t_2t_3]$.

The characteristics equation of matrix A can be written by

$$\begin{aligned}&\omega^3-\omega^2(n_2+t_3+u_1)\\&+\omega(n_2t_3-n_3t_2+n_2u_1+t_3u_1-n_1u_2-t_1u_3)\\&+n_3t_2u_1-n_2t_3u_1-n_3t_1u_2+n_1t_3u_2+n_2t_1u_3-n_1t_2u_3\\&=0\end{aligned} \tag{35}$$

The eigenvalue of matrix A are the three roots of eqs. (35) which define by the forms $\omega_1, \omega_2, \omega_3$. Thus, the eigenvectors Y are computed as:

$$Y=\begin{pmatrix}-n_3u_2+(-\omega+n_2)u_3\\ n_3(-\omega+u_1)-n_1u_3\\ -\omega^2+n_2(\omega-u_1)+\omega u_1+n_1u_2\end{pmatrix}. \tag{36}$$

The solutions of eqs. (34) which are bounded as $r\to\infty$ can be given by

$$\mathrm{V}(r,p)=\sum_{i=1}^{3}R_iY_iK_1(n_ir) \tag{37}$$

where $n_i=\sqrt{\omega_i}$, R_1, $R_2$and $R_3$are constants can be determined by using the problem boundary conditions and $K_1$is the order one modified Bessel's function. Thus, the solutions of physical queitites depend on r and p can be taken the forms:

$$\overline{u}(r,p)=\sum_{i=1}^{3}R_iU_iK_1(n_ir) \tag{38}$$

$$\overline{N}(r,p)=-\sum_{i=1}^{3}R_i\frac{N_i}{n_i}K_0(n_ir) \tag{39}$$

$$\overline{\phi}(r,p)=-\sum_{i=1}^{3}R_i\frac{T_i}{n_i}K_0(n_ir) \tag{40}$$

$$\overline{T}(r,p)=-\sum_{i=1}^{3}R_i\frac{T_i\left(1-bn_i^2\right)}{n_i}K_0(n_ir) \tag{41}$$

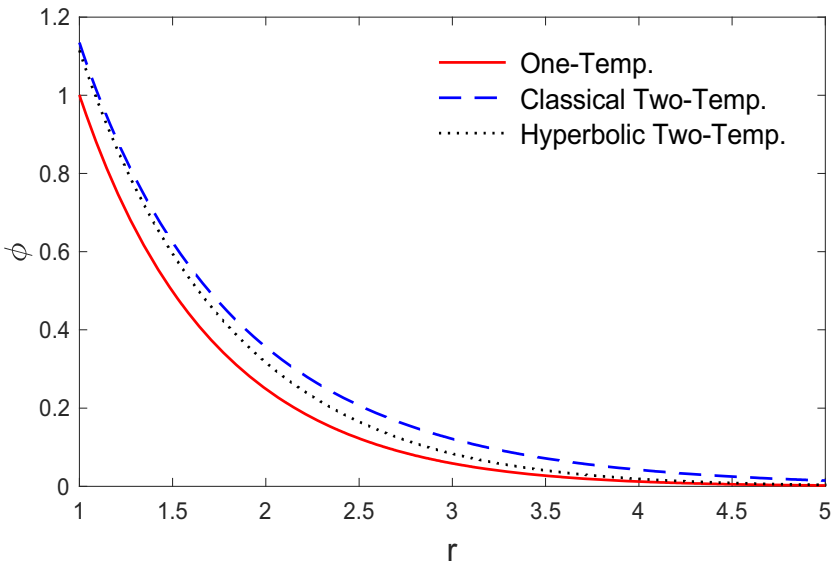


Fig. 1 The variation of conductive temperature with distance for different models

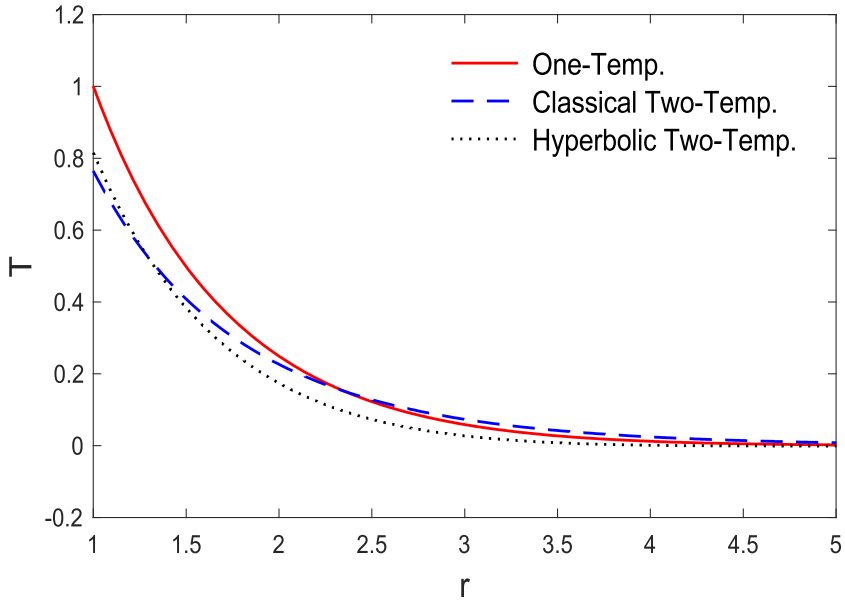


Fig. 2 The variation of thermodynamic temperature with distance for different models

$$\overline{\sigma}_{rr}(r,p) = \sum_{i=1}^{3} R_i \left(\frac{-n_i^2 U_i + \epsilon_1 N_i + \left(1-bn_i^2\right) r_2 T_i}{n_i} K_0(n_i r) + \frac{(r_7-1)U_i}{r} K_1(n_i r) \right) \tag{42}$$

$$\overline{\sigma}_{\theta\theta}(r,p) = \sum_{i=1}^{3} R_i \left(\frac{-r_7 n_i^2 U_i + \epsilon_1 N_i + \left(1-bn_i^2\right) r_2 T_i}{n_i} K_0(n_i r) - \frac{(r_7-1)U_i}{r} K_1(n_i r) \right) \tag{43}$$

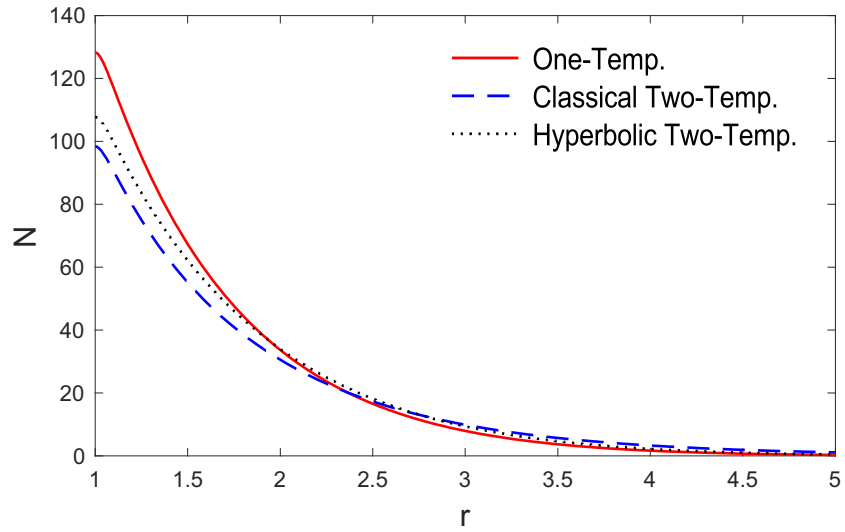


Fig. 3 The variation of carrier density with distance for different models

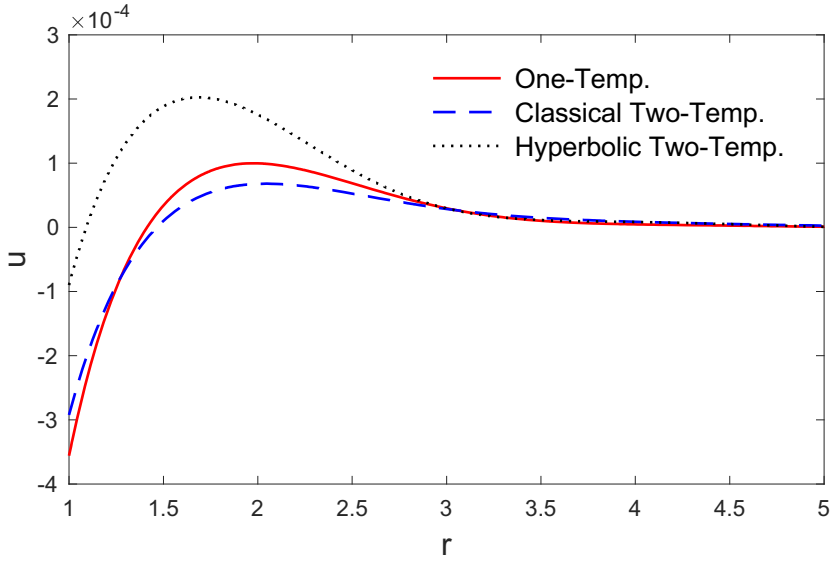


Fig. 4 The variation of displacement with distance for different models

The numerical inversions scheme adopted the general solutions of the radial displacement, the carrier density, the conductive and thermodynamic temperatures and the stresses distribution. The Stehfest scheme [38] was chosen as a numerical inversion method. In this approach, the Laplace transform inverse for $\overline{\Omega}(r,p)$ are approximated as

$$\Omega(r,t) = \frac{ln(2)}{t} \sum_{n=1}^{N} V_n \overline{\Omega}\left(r, n\frac{ln(2)}{t}\right) \tag{44}$$

where

$$V_n = (-1)^{\left(\frac{N}{2}+1\right)} \sum_{p=\frac{n+1}{2}}^{min\left(n,\frac{N}{2}\right)} \frac{(2p)!p^{\left(\frac{N}{2}+1\right)}}{p!(n-p)!\left(\frac{N}{2}-p\right)!(2n-1)!},$$

where N is the term numbers.

5 Numerical Results and Discussions

Now, we consider a numerical example for the calculations purpose similar to the silicon (Si) medium with the physical constants as in [39]:

$c_e = 695(J)(kg^{-1})(k^{-1})$, $\rho = 2330(kg)(m^{-3})$, $E_g = 1.11\ (eV)$, $t_p = 2(ps)$,

$d_n = -9 \times 10^{-31}(m^3), \lambda = 3.64 \times 10^{10}(N)(m^{-2})$, $\alpha_t = 3 \times 10^{-6}(k^{-1})$,

$\mu = 5.46 \times 10^{10}(N)(m^{-2}), \tau = 5 \times 10^{-5}(s), s_o = 2\ (m)(s^{-1}), n_o = 10^{20}(m^{-3})$,

$D_e = 2.5 \times 10^{-3}(m^2)(s^{-1}), T_o = 300(k)$.

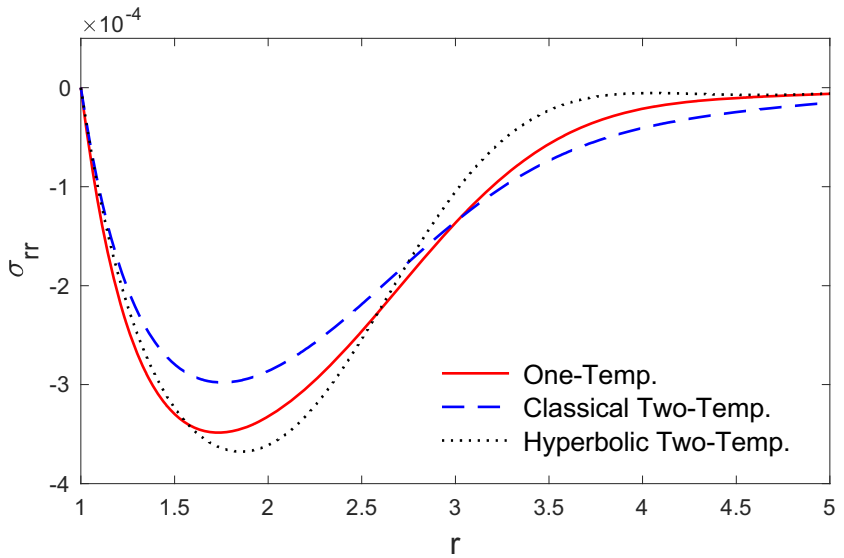


Fig. 5 The variation of radial stress with distance for different models

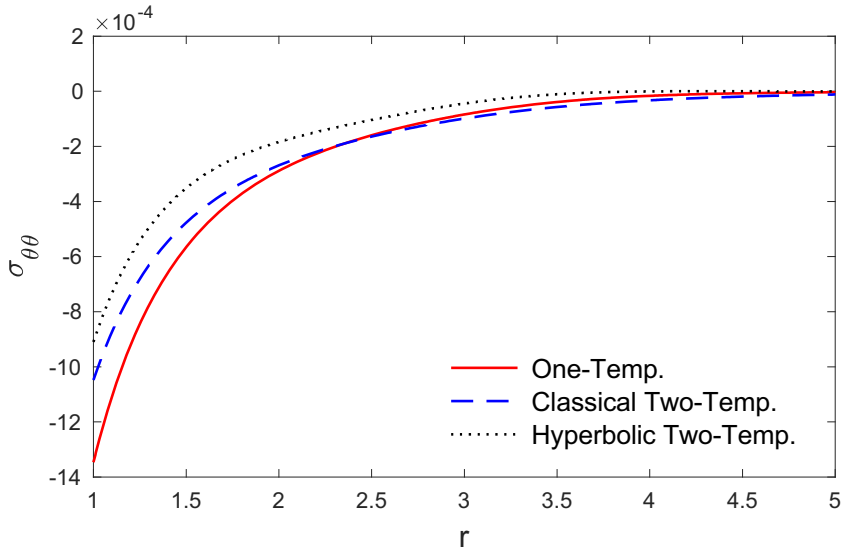


Fig. 6 The variation of hoop stress with distance for different models

The calculations are carried out for the time $t = 2.0$. Based on the above constants list, the calculated physical quantities (numeral) along the radial distance r with one temperature, classical two-temperature and hyperbolic two-temperature are represented in figs. 1–6. The variations of carrier density, the variations of radial displacement, the variations of conductive temperature, the variations of thermodynamic temperature, the radial and hoop stresses distributions with respect to the redial distance r in the context of the coupled photo-thermal models are numerically computed. Figure 1 shows the variation of conductive temperature with respect to the redial distance r. It is noticed that the conductive temperature has ultimate value at the boundary $r = 1$ then it gradually decreases with the increasing of the radial distance r till it reaches to zeros. Figure 2 explains the variation of thermodynamic temperature along the radial distance r. It is noticed that all curves start from the maximum value at the boundary $r = 1$ and reduces with the raising of the redial distance r to come to zeros for sufficiently large values of r. Fig. 3 exhibits the variations of carrier density versus the radial distance r. It was shown that the carrier density starts with its maximum value at the boundary $r = 1$ after that it gradually decreases with the increasing of radial distance r till it close to zeros. Figure 4 displays the variation of radial displacement with respect to r. It is noticed that the displacement attains maximum negative values after that it increases gradually up to peak value after that decreases to zeros. Figure 5 displays the variation of radial stress along the radial distance r. It can be observed that the radial stress, always starts from zeros and drop to zeros to obey the boundary condition. The variations of hoop stress with respect to r was displayed in fig. 6. It is clear that the hoop stress attains utmost negative values after that constantly increases to zeros. Finally, in the compressions between the solutions, one can conclude that considering the coupled photo-thermal model with a new hyperbolic two-temperature is a significant phenomenon and have major effects on the physical quantities distributions.

6 Conclusions

Based on this hyperbolic two-temperature model of thermo-elasticity, the variation of carrier density, the variations of thermodynamic temperature and the conductive temperature, the variations of radial displacement, the stress components distributions in semiconducting media are presented. In comparison between the solutions, one can conclude that considering the coupled photo-thermo-elastic model with a new hyperbolic thermal temperature and conductive temperature are a great phenomenon and have major effects on field quantities distribution.

Acknowledgments This project was funded by the Deanship of Scientific Research (DSR) at King Abdulaziz University, Jeddah, Saudi Arabia, under grant no. (KEP-67-130-38). The authors, therefore, acknowledge with thanks DSR technical and financial support.

References

1. Gurtin ME, Williams WO (1967) An axiomatic foundation for continuum thermodynamics. Arch Ration Mech Anal 26(2):83–117
2. Chen PJ, Gurtin ME (1968) On a theory of heat conduction involving two temperatures. Zeitschrift für angewandte Mathematik und Physik (ZAMP) 19(4):614–627
3. Chen PJ, Gurtin ME, Williams WO (1969) On the thermodynamics of non-simple elastic materials with two temperatures. Zeitschrift für angewandte Mathematik und Physik ZAMP 20(1):107–112
4. Youssef HM, El-Bary AA (2018) Theory of Hyperbolic Two-Temperature Generalized Thermoelasticity. Mater Phys Mech 40: 158–171

5. Ezzat, MA, El-Karamany, AS, Ezzat, SM (2012) Two-temperature theory in magneto-thermoelasticity with fractional order dual-phase-lag heat transfer. Nuclear Engineering and Design, 252): 267–277
6. Ezzat MA, Youssef HM (2014) Two-temperature theory in three-dimensional problem for thermoelastic half space subjected to ramp type heating. Mech Adv Mater Struct 21(4):293–304
7. Deswal, S, Kalkal, KK (2013) Two temperature magneto-thermoelasticity with initial stress: State space formulation. Journal of Thermodynamics, 1(1
8. Abbas IA, Kumar R, Reen LS (2014) Response of thermal source in transversely isotropic thermoelastic materials without energy dissipation and with two temperatures. Can J Phys 92(11):1305–1311
9. Todorović D (2003) Plasma, thermal, and elastic waves in semiconductors. Rev Sci Instrum 74(1):582–585
10. Todorović D (2003) Photothermal and electronic elastic effects in microelectromechanical structures. Rev Sci Instrum 74(1):578–581
11. Marin M, Vlase S, Ellahi R, Bhatti M (2019) On the partition of energies for the backward in time problem of thermoelastic materials with a dipolar structure. Symmetry 11(7):863
12. Marin M, Ellahi R, Chirilă A (2017) On solutions of saint-Venant's problem for elastic dipolar bodies with voids. Carpathian J Mathematics 33(2):219–232
13. Song Y, Todorovic DM, Cretin B, Vairac P (2010) Study on the generalized thermoelastic vibration of the optically excited semiconducting microcantilevers. Int J Solids Struct 47(14–15):1871–1875
14. Othman MI, Marin M (2017) Effect of thermal loading due to laser pulse on thermoelastic porous medium under GN theory. Results in Physics 7:3863–3872
15. Yasein M, Mabrouk N, Lotfy K, EL-Bary A (2019) The influence of variable thermal conductivity of semiconductor elastic medium during photothermal excitation subjected to thermal ramp type. Results in Physics 15:102766
16. Ailawalia, P, Kumar, A (2019) Ramp Type Heating in a Semiconductor Medium under Photothermal Theory. Silicon): 1–10
17. Alzahrani FS, Abbas IA (2019) Photo-thermo-elastic interactions without energy dissipation in a semiconductor half-space. Results in Physics 15:102805
18. Lotfy K, El-Bary A, Hassan W, Alharbi A, Almatrafi M (2020) Electromagnetic and Thomson effects during photothermal transport process of a rotator semiconductor medium under hydrostatic initial stress. Results in Physics 16:102983
19. Lotfy K, Hassan W, El-Bary A, Kadry MA (2020) Response of electromagnetic and Thomson effect of semiconductor medium due to laser pulses and thermal memories during photothermal excitation. Results in Physics 16:102877
20. Lotfy K, El-Bary A, Tantawi R (2019) Effects of variable thermal conductivity of a small semiconductor cavity through the fractional order heat-magneto-photothermal theory. European Phys J Plus 134(6):280
21. Hobiny A, Abbas I (2019) A GN model on photothermal interactions in a two-dimensions semiconductor half space. Results in Physics 15:102588
22. Ali, H, Jahangir, A, Khan, A (2019) Reflection of waves in a rotating semiconductor nanostructure medium through torsion-free boundary condition. Indian J Phys: 1–9
23. Jahangir A, Tanvir F, Zenkour A (2020) Photo-thermo-elastic interaction in a semiconductor material with two relaxation times by a focused laser beam. Advances Aircraft Spacecraft Sci 7(1):41–52
24. Abbas IA (2014) The effects of relaxation times and a moving heat source on a two-temperature generalized thermoelastic thin slim strip. Can J Phys 93(5):585–590
25. Zenkour, AM, Abbas, IA (2014) A generalized thermoelasticity problem of an annular cylinder with temperature-dependent density and material properties. International Journal of Mechanical Sciences, 84): 54–60
26. Abbas IA (2014) Nonlinear transient thermal stress analysis of thick-walled FGM cylinder with temperature-dependent material properties. Meccanica 49(7):1697–1708
27. Lotfy K (2019) Effect of variable thermal conductivity during the photothermal diffusion process of semiconductor medium. Silicon 11(4):1863–1873
28. Lotfy, K, Abo-Dahab, S, Tantawy, R, Anwar, N (2019) Thermomechanical Response Model on a Reflection Photothermal Diffusion Waves (RPTD) for Semiconductor Medium. Silicon): 1–11
29. Hobiny, A, Abbas, I (2019) Fractional Order GN Model on Photo-Thermal Interaction in a Semiconductor Plane. Silicon): 1–8
30. Abbas IA, Aly K, Dahshan A (2018) Analytical solutions of plasma and Thermoelastic waves Photogenerated by a focused laser beam in a semiconductor material. Silicon 10(6):2609–2616
31. Song Y, Bai J, Ren Z (2012) Study on the reflection of photothermal waves in a semiconducting medium under generalized thermoelastic theory. Acta Mech 223(7):1545–1557
32. Mandelis A, Nestoros M, Christofides C (1997) Thermoelectronic-wave coupling in laser photothermal theory of semiconductors at elevated temperatures. Opt Eng 36(2):459–468
33. Zenkour AM, Abouelregal AE (2015) Effect of temperature dependency on constrained orthotropic unbounded body with a cylindrical cavity due to pulse heat flux. Journal of Thermal Science and Technology 10(1):JTST0019–JTST0019
34. Das NC, Lahiri A, Giri RR (1997) Eigenvalue approach to generalized thermoelasticity. Indian J Pure Appl Mathematics 28(12): 1573–1594
35. Abbas IA (2014) Eigenvalue approach for an unbounded medium with a spherical cavity based upon two-temperature generalized thermoelastic theory. J Mech Sci Technol 28(10):4193–4198
36. Abbas IA (2015) A dual phase lag model on thermoelastic interaction in an infinite fiber-reinforced anisotropic medium with a circular hole. Mechanics Based Design Struct Machines 43(4):501–513
37. Abbas IA (2015) The effects of relaxation times and a moving heat source on a two-temperature generalized thermoelastic thin slim strip. Can J Phys 93(5):585–590
38. Stehfest H (1970) Algorithm 368: numerical inversion of Laplace transforms [D5]. Commun ACM 13(1):47–49
39. Song Y, Todorovic DM, Cretin B, Vairac P, Xu J, Bai J (2014) Bending of semiconducting cantilevers under Photothermal excitation. Int J Thermophys 35(2):305–319

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